Content-Based Vocal Repertoire Ranking Framework Using Duration-Weighted Pitch Distributions

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Abstract

Choosing vocal repertoire that fits a singer's voice is important for vocal health, yet to the author's knowledge, current tools based on tessituragrams help assess a piece only after it has been selected—they do not recommend new pieces likely to suit the singer. A key factor in vocal fit is tessitura: the pitches on which the voice spends the most time in a given piece, weighted by how long each pitch is sustained. This paper presents a proof-of-concept content-based ranking framework that uses duration-weighted tessituragrams—pitch-by-duration profiles extracted from machine-readable musical scores—to rank songs by how well they match a singer's stated preferences. The singer specifies a comfortable vocal range, favorite pitches, and pitches to avoid; the system filters songs by range, constructs an ideal pitch profile from these preferences, and scores each candidate using a similarity measure (cosine similarity) that captures how closely the song's pitch profile aligns with the ideal, minus a penalty for time spent on pitches the singer wants to avoid. We evaluate offline with synthetic self-retrieval: a profile is built from a target vocal line (the range of the vocal line, four pitches with the most total duration as favorites, and the two with the least as avoids), and we ask how often that same line ranks highly among other filtered lines. We evaluate two protocols on the OpenScore Lieder corpus: a compact library of 101 vocal lines, one per composition, and an expanded flattened library of 1,655 vocal lines drawn from 1,419 compositions (some works contribute more than one line). We compare the full model to two different baselines: null (range filter and random ranking) and cosine-only (without an avoid penalty). The full model hit rate at 1 (HR@1) is 76% vs 6% null (compact design) and 55% vs 2% (expanded); HR@5 is 86% vs 7% null (expanded). Rankings are stable under small preference edits (mean Kendall's $\tau \approx 0.84$-0.85). The avoid penalty does not clearly improve HR@1 or mean reciprocal rank over cosine-only on these offline draws. Results support internal consistency under these offline protocols. Claims are limited to synthetic profiles and symbolic scores; human evaluation and other genres are future work.

Keywords: tessitura, tessituragrams, vocal repertoire recommendation, content-based recommendation, cosine similarity, music information retrieval, MusicXML

Introduction

Choosing repertoire that matches a singer's vocal characteristics is important for vocal health: misalignment between a piece's demands and a singer's capabilities raises injury risk (Apfelbach, 2022). A key demand is tessitura—where the line sits in range, i.e. which pitches receive the most singing time (Apfelbach, 2022; Schloneger et al., 2024; Thurmer, 1988). In practice, singers and teachers often filter by a piece's written range (the highest and lowest notes in the score) or by vocal classification such as Fach—the system used in opera to categorize voices by range, weight, and color. A piece's written range alone, however, does not show how much time the voice spends in different parts of that span, and Fach labels are not consistently defined across pedagogical contexts (Schloneger et al., 2024).

A tessituragram histograms singing time per notated pitch, typically duration-weighted so longer notes count more (Thurmer, 1988; Titze, 2008). Tools such as Tessa (Apfelbach, 2022) extract tessituragrams from MusicXML and report summaries. Those tools are analytic: they help judge a piece after it is chosen, not find new pieces whose tessitura is likely to fit.

We ask whether tessituragrams can rank repertoire from content. We build a range-filtered pipeline: an ideal profile from favorite and avoid pitches, cosine similarity to candidates, and a penalty for time on avoided pitches. Two offline experiments use synthetic user profiles—different protocols, not repeated runs for weight. Experiment 1: 101 lines (one per work), compact library, independent and identically distributed (i.i.d. bootstrap). Experiment 2: 1,655 lines from 1,419 works (~16× more lines), flattened multi-part lines, larger post-filter pools, cluster bootstrap where dependence requires it. Claims are limited to these metrics and protocols, not clinical outcomes or unconstrained deployment.

Contributions: (1) A proof-of-concept ranking pipeline combining duration-weighted tessituragrams with user-specified favorite pitches and avoid pitches, scored using cosine similarity minus an avoid-penalty. (2) A two-protocol synthetic self-retrieval plus stability and implementation checks vs null and cosine-only baselines. (3) An offline evaluation protocol for the expanded library that samples queries over eligible vocal lines and uses work-level (cluster) bootstrap inference where multiple lines can come from the same composition (Cameron et al., 2008; Field and Welsh, 2007).

Related Work

Tessituragram Analysis and Repertoire Selection

Thurmer (1988) formalized the tessituragram; Titze (2008) emphasized duration weighting. Recent work uses duration- and dose-based summaries across works or cycles (Schloneger et al., 2024; Patinka, 2024). Nix (2014) evaluates objective methods—including tessituragrams—for matching pieces to singers; Apfelbach's Tessa (2022) automates extraction from MusicXML. That line of work supports evaluating a selected piece, not recommending pieces whose tessitura is likely to fit.

Music Information Retrieval

Music information retrieval (MIR) addresses content-based search and retrieval of music, including symbolic (score-based) methods (Casey et al., 2008; Gurjar and Moon, 2018). For symbolic music, pitch histograms and duration distributions are common features (Corrêa and Rodrigues, 2016). Cosine similarity suits such pitch-indexed vectors because it measures proportional alignment (Müller, 2015). For the voice, octave matters: unlike transposition-invariant melodic similarity (Mongeau and Sankoff, 1990), tessitura modelling must keep pitch in specific octaves to reflect register and demand. Our system is related to the query-by-example paradigm in MIR (retrieving items similar to a user-provided example), though the user supplies pitch preferences rather than an example song; these preferences are matched against score-derived tessituragrams (Casey et al., 2008; Müller, 2015).

Recommender Evaluation

Offline evaluation is standard without human relevance labels (Herlocker et al., 2004; Urbano et al., 2013). In music recommendation, offline metrics can align with later A/B tests (Gruson et al., 2019); shared metric code improves comparability (Raffel et al., 2014). We report hit rate at k (HR@k), mean reciprocal rank (MRR), Kendall's τ (Kendall, 1948), and bootstrap uncertainty (Efron and Tibshirani, 1993). When several vocal lines can come from one composition, cluster bootstrap resamples works, not lines (Cameron et al., 2008; Field and Welsh, 2007). We are not aware of prior tessitura-based vocal recommenders evaluated with these metrics and explicit baselines. Our evaluation follows these practices and is described in the following section.

Method

Data and Song Library

All sources come from the OpenScore Lieder Corpus (CC0; Gotham and Jonas, 2022)—predominantly Lieder (German art song) and French mélodie (the French counterpart). We parse MusicXML with music21 (Cuthbert and Ariza 2010), extract the vocal line, and build one tessituragram per line: a mapping from each MIDI pitch (e.g. 60 = middle C) to total duration in quarter-note beats—MIDI assigns a number to each pitch so octaves are explicit. Duration weighting reflects that sustaining a pitch is more demanding than a brief note (Titze 2008). Table 1 lists features extracted or derived per vocal line.

Two libraries are used. Experiment 1 (Exp 1) contains 101 art songs. Experiment 2 (Exp 2) contains 1,655 vocal lines from 1,419 compositions. Because some compositions contain multiple voice parts (e.g. duets), each vocal line is treated separately. No human relevance judgements were collected.

We report two protocols, not two independent replications. Exp 1 uses a compact single-line-per-work library and i.i.d. bootstrap over query-level outcomes. Exp 2 uses a flattened line set and work-level (cluster) bootstrap when sampled queries share a composition. The libraries overlap in source and repertoire; they are not two independent draws from disjoint populations, so they should not be read as doubling evidence.

Feature	Source	Description
Pitch, duration (per note)	MusicXML	Converted to MIDI number; duration in quarter beats, summed per pitch for tessituragram.
Part/voice	MusicXML	Vocal line only.
Tessituragram	Derived	MIDI pitch → total duration (duration-weighted).
min_midi, max_midi	Derived	Written pitch range of the vocal part.
Composer, title, filename	MusicXML/corpus	Work and file identification.

Table 1. Features Extracted from MusicXML and Stored per Vocal Line

Ranking Framework and Scoring

Inputs: The singer supplies (1) a comfortable vocal range (lowest and highest MIDI pitch); (2) favorite notes; (3) notes to avoid.

Range filter: Songs whose written pitch range extends beyond the singer's stated range are excluded, so every retained candidate is singable within that span (candidate set C).

Ideal vector construction: A dense vector over [minmidi, ..., maxmidi] is initialized to a base weight of 0.2, with +1.0 at favorites and −1.0 at avoids. The base weight keeps non-favorite in-range pitches from being zeroed out before normalization; we do not sweep this hyperparameter in our experiments. Negative entries are clamped to 0 and the vector is L2-normalized so similarity depends only on direction (Müller, 2015).

Song scoring and ranking: Each song's tessituragram is converted to a dense vector and L1-normalised so each entry is the proportion of total vocal duration on that pitch; the avoid penalty is therefore the total mass on avoided pitches. The final score is:

$$final_score = cosine_similarity(song, ideal) - \alpha \times avoid_penalty$$

where α controls the weight of the avoid penalty (α = 0.5 in the main experiments). The avoid penalty is the proportion of the song's vocal duration spent on notes to avoid. Scores are rounded to four decimal places; ties are broken by filename.

Synthetic User Profiles and Statistical Methods

Synthetic ideal profile: For a profile line, the 'user's' range equals that line's written range; favorites are the top 4 pitches by duration and avoids the bottom 2, disjoint. This fixed sparse profile rule is a test harness—not claimed psychologically realistic—but it yields a reproducible identifiability target. Synthetic self-retrieval asks whether the ranker recovers that line among range-filtered others, not whether human judges would agree (Herlocker et al., 2004; Urbano et al., 2013).

Exp 1 (101-line library): 50 queries uniformly without replacement from the valid pool (|C| ≥ 2; seed 42). 10,000 i.i.d. bootstrap resamples of query- (or baseline-) level outcomes.

Exp 2 (1,655-line library): Up to 200 queries uniformly without replacement from eligible lines (not compositions). Pieces with more than one vocal line (e.g. duets) therefore show up more often in the query list than they would if we sampled works with equal probability. For confidence intervals on query-level metrics (RQ1), we use work-level (cluster) bootstrap (Cameron et al., 2008; Field and Welsh, 2007): we resample compositions with replacement and keep all sampled query lines from each chosen work. That matches the fact that several queries can come from the same piece. Seed 42 fixes which queries are drawn; seed 43 fixes the bootstrap resampling so the two steps stay separately reproducible.

Inference scope: Bootstrap percentile intervals summarize uncertainty for the mean query-level metric on the fixed sampled queries; they are not intervals over redrawn query sets. RQ3 pools Spearman correlations via Fisher's z (Fisher, 1915). RQ1 main results and baselines use the same query draw. HR@1 and MRR are primary metrics for self-retrieval; HR@3/5 are complementary. Paired full vs cosine-only uses the same queries; we do not apply multiplicity correction across metrics—secondary contrasts are exploratory.

Results

We report RQ1 (self-retrieval), RQ2 (stability), α-sensitivity, and RQ3 (implementation). Tabulated values reproduce the canonical JSON outputs for the stated seeds and paths (Method).

Research Question 1: Self-Retrieval Accuracy

Question: If preferences are built from a vocal line's own tessituragram, does the system rank that line first (HR@1), or in the top 3 (HR@3), or 5 (HR@5)?

The profile uses a coarse summary (top-4 favorite and bottom-2 avoid MIDI pitches). RQ1 tests whether the scoring pipeline recovers the generating item among other range-filtered candidates under this fixed sparse encoding—a pipeline identifiability check—whether the scorer recovers the generating item among distractors.

Figure 1 shows how often the correct song is ranked first ("hit rate at 1") and average rank quality (higher = the correct song tends to appear closer to the top). Parentheses are uncertainty ranges for the average on this one batch of test queries.

Summary: In the smaller song list, the full model ranks the true song first 76% of the time, cosine-only 80%, and random ordering 6% (roughly what chance would give after filtering). On the same queries, cosine is a few points ahead of the full model on first place; that gap is modest. In the larger list, first-place rates are about 55% for both models—effectively tied—and still far above random (2%). The two tests do not isolate "big library versus small": they use different song sets, different numbers of

competitors after filtering, and different ways of picking test songs—so do not read the drop from 76% to 55% as proof that a bigger catalog alone caused it. Where many songs look alike, the true song will often land lower; in the larger test, it is not first about 45% of the time.

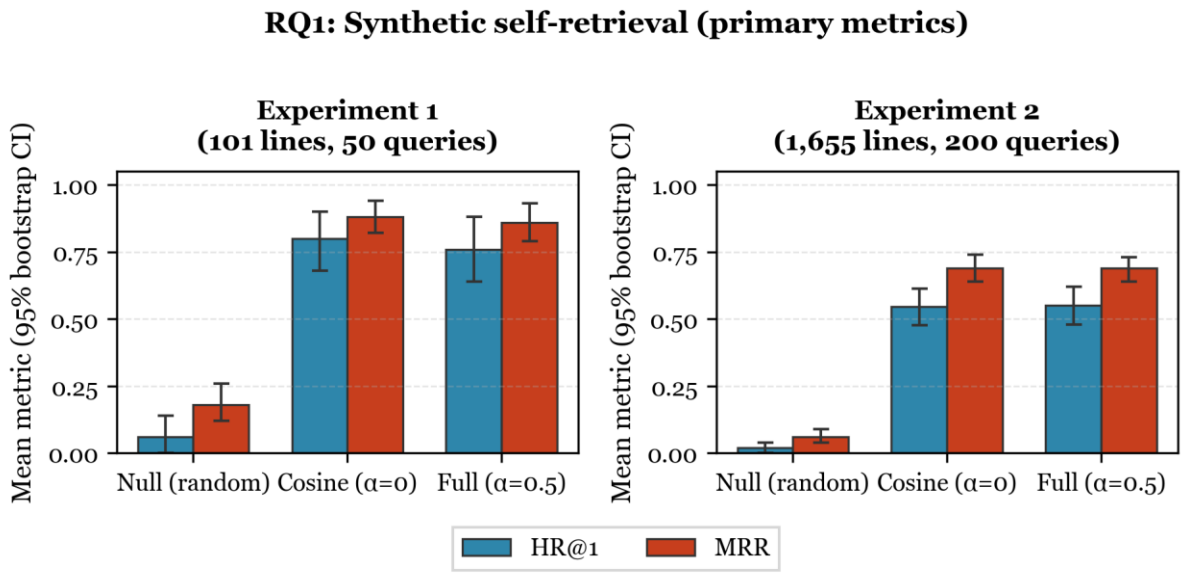

Figure 1. How often the target is ranked first (HR@1) and average rank quality (MRR), with 95% intervals for this sample (Method). Left: Experiment 1 (smaller library; i.i.d. bootstrap). Right: Experiment 2 (larger library; cluster bootstrap). The near-tie full vs cosine first-place rates in Experiment 2 are 55.0% vs 54.5%.

Research Question 2: Ranking Stability

Question: When we add or remove one favorite note or note to avoid, how similar is the new ranking to the original?

Procedure: In Experiment 1, we use 5 baseline profiles (candidate set ≥ 10 songs each, 130 total perturbations). In Experiment 2, we use 20 baseline profiles (candidate set ≥ 10, 580 total perturbations, all 20 from distinct compositions). For each baseline, we obtain the reference ranking and generate all one-note perturbations (add or remove one favorite or avoid note). For each perturbation we compute Kendall's τ (Kendall, 1948). τ ranges from −1 to 1; values above 0.7 indicate strong agreement. Table 2 gives mean τ per baseline with 95% CI. τ measures how much the induced ranking moves under local edits to the encoded preference lists; it does not validate pedagogical acceptance or user satisfaction with such edits.

Both experiments show strong stability (τ > 0.7). Mean τ for the full model is 0.85 (Experiment 1) and 0.84 (Experiment 2)—similar despite about 16× more vocal lines in Experiment 2 and four times as many stability baselines. Cosine-only is slightly higher in both cases (0.87), with overlapping CIs. The null baseline hovers near τ ≈ 0 as expected.

Model	Exp 1:	Exp 2:
	Mean τ (95% CI)	Mean τ (95% CI)
Null (random)	−0.04 [−0.05, −0.02]	0.00 [-0.002, 0.01]
Cosine-only (α = 0)	0.87 [0.84, 0.91]	0.87 [0.86, 0.88]
Full (α = 0.5)	0.85 [0.81, 0.88]	0.84 [0.82, 0.85]

Table 2. Ranking stability: mean Kendall's τ per baseline (95% CI). τ = 1 means identical order; τ near 0 means unrelated rankings.

Sensitivity to α

α controls how strongly the scorer penalizes time spent on notes to avoid (α = 0 is cosine-only—the penalty is off). We repeat RQ1 (self-retrieval) and RQ2 (stability) at α ∈ {0, 0.25, 0.5, 0.75, 1}, using the same query draws and RQ2 baselines as Figure 1 and Table 2 (Method).

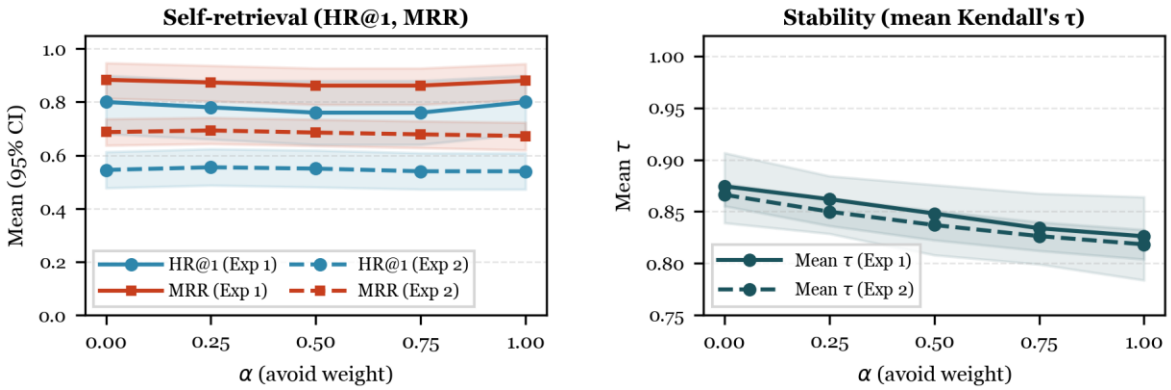

Figure 2. Vocal lines = mean metric on the fixed sample; shaded bands = 95% bootstrap uncertainty. Rows: Experiment 1 (top), Experiment 2 (bottom). Left: self-retrieval—HR@1 is how often the true line is ranked first; MRR summarizes how high it ranks on average. Right: stability—τ measures how similar the ranking stays after small edits to favorites/avoids (higher = more agreement; dashed line at 0.7 marks "strong" agreement). HR@3 and HR@5 are not plotted; point estimates vary little with α (Exp 1: HR@3 0.96–1.00, HR@5 1.00; Exp 2: HR@3 0.78–0.81, HR@5 0.84–0.86); full CIs in archived JSON (Method).

Research Question 3 (RQ3): Implementation

As a verification step, we check three properties of the scoring pipeline in both experiments: (a) Do scores spread out meaningfully? (b) Does finalscore = cos − α × avoid hold numerically? (c) Do components correlate in the directions the formula predicts?

In Experiment 1, we use 25 synthetic profiles (≥ 10 candidates each). In Experiment 2, we use 50 profiles (≥ 10 candidates each, 49 from distinct compositions). Table 3 reports the identity residual max final − (cos − 0.5×avoid), OLS recovery of the linear weights, mean range of finalscore, and Spearman ρ (Spearman, 1904) aggregated across profiles within each experiment via Fisher's z (Fisher, 1915).

Quantity	Experiment 1 (25 profiles)	Experiment 2 (50 profiles)
max |final − (cos − 0.5×avoid)|	0	0
Mean range (finalscore)	0.758	1.030
ρ(final, cos)	0.987	0.989
ρ(final, avoid)	-0.350	-0.318
ρ(cos, favoriteoverlap)	0.935	0.921

Table 3. RQ3 implementation checks. Identity residual is pointwise finalscore − (cos − 0.5×avoid) at α = 0.5. Spearman ρ aggregated across profiles via Fisher's z.

Discussion and Limitations

Range filtering plus cosine similarity over duration-weighted pitch profiles strongly beats random ordering after the same filter and stays stable under small preference edits in both protocols. Experiment 2 adds larger |C| and 1,655 lines from 1,419 compositions. Key quantities are consistent in direction across protocols (not independent confirmations): mean τ for the full model is 0.85 (Experiment 1) and 0.84 (Experiment 2); RQ3 confirms the implemented score matches cos − α×avoid; full and cosine-only dominate the null on self-retrieval.

Raw HR@1 falls from 0.76 to 0.55 (full model) between experiments, but—as in the Results—this gap confounds corpus composition, |C|, null difficulty, and line-level sampling weights; it is not identified with library size alone. In Experiment 2, HR@5 for the full model is 0.86.

On the primary RQ1 metrics we emphasize (HR@1, MRR), cosine-only and the full model are not separated by paired intervals in Experiment 2, and Experiment 1's paired ΔHR@1 favors cosine-only on this draw. Choosing between them is therefore not dictated by these offline self-retrieval results alone. The avoid term remains part of the design when users supply pitches to avoid; our evaluation rule defines those avoids from the least-used pitches of the target line, which limits how much the penalty can separate the target from near-duplicate lines—so paired full-versus-cosine contrasts here mainly probe cosine similarity, not a high-signal test of user-supplied avoids. Bootstrap schemes differ only where Experiment 2 requires cluster resampling (Method).

Limitations: (1) Scope of inference. We use one fixed random query sample (seed 42) on one corpus; bootstrap intervals describe uncertainty for that sample, not for all music or new query draws. In Experiment 2, lines from the same work can be related. Contrasts between full and cosine-only models do not adjust for multiple metrics. (2) No human ratings. Profiles are synthetic; RQ1 asks whether the system recovers a hidden target line under a fixed rule—not whether real singers would like the rankings or hold realistic preferences. (3) Comparing experiments. Side-by-side numbers mix different library sizes, candidate pools, and sampling rules; they are descriptive, not proof of a causal effect of catalog size, and the two protocols are not independent replications. (4) Corpus. Both runs use the OpenScore Lieder art-song corpus; other genres (e.g. opera, popular song) are untested. (5) What we model. Only pitch and time spent on each pitch; we omit dynamics, tempo, text, and accompaniment. (6) Full vs cosine-only. Paired intervals for HR@1 and MRR often overlap (Experiment 2 includes zero; Experiment 1's ΔHR@1 favors cosine on this draw). Because avoids are inferred from the target line itself, full-versus-cosine contrasts do not isolate user-chosen avoids; they mainly reflect cosine similarity.

Conclusion

We presented a content-based ranking framework that uses duration-weighted tessituragrams to rank vocal repertoire by pitch-preference match. Across two offline protocols—compact single-line-per-work library (101 lines) and expanded flattened library (1,655 lines from 1,419 compositions)—the full model and cosine-only baseline strongly outperform a null baseline on synthetic self-retrieval, and rankings stay stable under small preference perturbations (mean τ ≥ 0.82 for the full model at every α in Figure 2). RQ3 confirms the score implements final = cos − α × avoid. Mean ranking stability is similar in both experiments (τ ≈ 0.85 vs. 0.84); in Experiment 2, HR@5 for the full model is 0.86. Paired contrasts on the primary self-retrieval metrics do not support a systematic mean advantage for the avoid penalty over cosine-only under under this synthetic self-retrieval evaluation.. Future work should evaluate with real user preferences and human relevance judgments, expand to more diverse repertoire (e.g. opera, musical theatre, popular song), and incorporate additional factors such as dynamics, tempo, and text setting.

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